

YAMPA VALLEY, CO, USA

WMO: 725710

Lat: 40.481N

Lon: 107.218W

Elev: 2012

StdP: 79.38

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94025

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-17.6	-26.1	0.4	-20.0	-22.0	0.7	-17.3	11.1	-0.5	9.3	-2.7	2.5	110	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	31.2	13.5	29.3	13.1	27.9	12.8	15.3	25.6	14.7	25.1	14.0	24.4	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.4	11.5	17.3	11.3	10.7	17.3	10.0	9.8	17.7	50.2	25.7	48.0	25.1	46.0	24.5	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	8.8	7.7	-25.9	33.1	3.1	1.3	-28.2	34.1	-30.0	34.9	-31.7	35.6	-34.0	36.6		
(5)			-26.0	16.4	2.9	1.0	-28.1	17.1	-29.8	17.7	-31.5	18.2	-33.6	19.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-6.9	-4.7	0.6	5.5	10.5	16.0	20.2	18.9	14.4	7.2	0.5	-6.2	(6)
(7)		DBStd	10.34	5.24	4.98	5.01	4.37	3.99	3.63	2.32	2.19	3.76	4.80	5.35	5.38	(7)
(8)		HDD10.0	2322	524	411	293	142	43	3	0	0	9	109	284	503	(8)
(9)		HDD18.3	4480	783	644	550	384	243	84	9	19	123	345	534	761	(9)
(10)		CDD10.0	1011	0	0	1	8	59	185	317	277	142	22	1	0	(10)
(11)		CDD18.3	126	0	0	0	0	1	15	68	37	6	0	0	0	(11)
(12)		CDH23.3	2297	0	0	0	1	31	434	1009	635	184	3	0	0	(12)
(13)		CDH26.7	672	0	0	0	0	3	117	358	166	29	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.1	3.1	3.4	3.9	3.8	3.9	3.7	3.7	3.7	3.5	3.3	3.1	(14)
(15)	Precipitation	PrecAvg	517	47	41	39	53	42	34	35	40	55	45	46	38	(15)
(16)		PrecMax	723	107	75	64	94	114	97	75	83	149	90	77	66	(16)
(17)		PrecMin	356	18	22	14	33	12	7	10	9	26	10	18	19	(17)
(18)		PrecStd	85	23	14	14	18	24	20	16	16	29	21	16	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	7.8	18.0	21.9	26.4	32.1	32.8	31.4	29.2	23.0	17.5	7.7	(19)
(20)			MCWB	1.1	2.2	6.3	8.5	11.3	13.4	14.1	13.5	12.7	9.3	6.3	2.1	(20)
(21)		2%	DB	3.0	5.9	14.0	18.9	23.8	29.3	32.0	30.0	27.4	21.2	14.0	4.1	(21)
(22)			MCWB	-0.2	1.1	4.6	7.1	10.3	12.4	13.9	13.1	11.7	8.8	4.7	0.4	(22)
(23)		5%	DB	2.1	3.7	11.3	17.2	22.2	27.7	30.2	28.5	25.9	18.7	11.9	2.6	(23)
(24)			MCWB	-0.4	0.2	3.7	6.2	9.7	12.1	13.5	12.8	11.1	7.9	4.1	-0.5	(24)
(25)		10%	DB	0.8	2.3	8.0	14.1	19.8	26.2	28.7	27.3	23.6	16.3	8.7	1.1	(25)
(26)			MCWB	-1.0	-0.5	2.3	5.1	9.1	11.7	13.4	12.6	10.5	6.6	2.8	-1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.8	6.8	8.9	12.5	14.7	16.4	15.8	14.6	10.8	7.0	2.6	(27)
(28)			MCDWB	4.4	7.3	16.9	19.8	23.9	26.9	26.3	24.7	24.7	20.0	15.6	7.2	(28)
(29)		2%	WB	0.6	1.5	5.3	7.5	11.2	13.8	15.6	15.1	13.1	9.5	5.4	0.9	(29)
(30)			MCDWB	2.6	4.9	13.4	17.3	21.6	26.1	26.0	24.5	22.8	18.8	13.1	3.8	(30)
(31)		5%	WB	-0.3	0.5	3.9	6.6	10.3	12.9	15.1	14.3	12.2	8.4	4.3	-0.1	(31)
(32)			MCDWB	1.6	3.4	10.4	15.8	20.1	25.4	25.7	23.9	21.5	17.3	11.0	2.2	(32)
(33)		10%	WB	-1.5	-0.3	2.6	5.6	9.3	12.2	14.4	13.7	11.4	7.3	2.9	-1.2	(33)
(34)			MCDWB	0.3	2.0	7.9	13.7	18.7	24.0	25.0	23.6	20.9	15.5	7.9	0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	9.2	11.0	12.5	13.9	16.7	16.4	15.8	15.4	13.6	11.3	9.0	(35)
(36)			MCDBR	9.2	10.2	15.1	16.8	17.7	19.1	18.4	18.1	18.4	17.9	15.8	9.8	(36)
(37)			MCWBR	7.0	7.0	7.9	7.9	7.8	7.7	6.5	6.6	7.5	8.7	8.6	6.9	(37)
(38)		5% WB	MCDBR	8.5	9.1	13.9	15.5	16.0	17.4	15.4	14.6	14.9	15.6	14.6	9.1	(38)
(39)			MCWBR	6.8	6.4	7.5	7.5	7.2	7.2	5.7	5.6	6.3	7.7	8.1	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.230	0.246	0.268	0.311	0.315	0.320	0.347	0.343	0.295	0.266	0.248	0.239	(40)	
(41)		taud	2.454	2.419	2.449	2.404	2.478	2.461	2.433	2.440	2.582	2.638	2.570	2.455	(41)	
(42)		Ebn at Noon	963	988	995	963	960	952	922	919	953	957	936	919	(42)	
(43)		Edh at Noon	75	91	100	113	108	110	112	108	88	73	67	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.22	3.05	4.43	5.46	6.31	7.59	7.07	6.10	5.18	3.63	2.43	1.84	(44)	
(45)		RadStd	0.14	0.22	0.41	0.39	0.47	0.46	0.30	0.27	0.33	0.30	0.27	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page